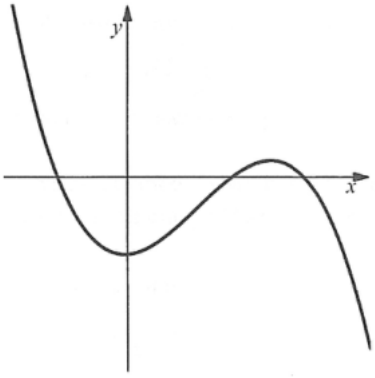
Differentiation: Points of Inflection and Applications

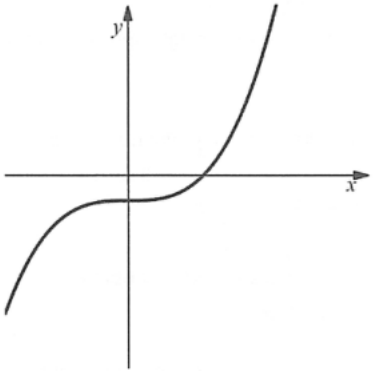
Questions 1–20

Question 1. For each of the following graphs, identify the points of inflection on a copy of the graph and describe the change in concavity at each point of inflection:

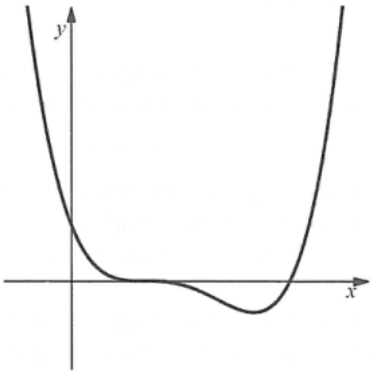
a)



b)



c)



Question 2. Sketch a graph with a single point of inflection at $(0, 0)$ which is non-stationary, concave (concave downwards) before $(0, 0)$ and convex (concave upwards) after $(0, 0)$

Question 3. Sketch a graph which has a stationary point which is a minimum at $x = -2$, a non-stationary point of inflection at $x = 1$ and a stationary point which is maximum at $x = 4$. Label each of the points clearly.

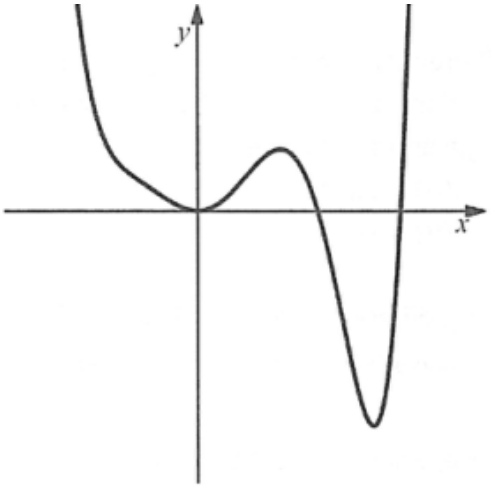
Question 4. Determine whether each of the curves are convex (concave upwards), concave (concave downwards) or neither at the given point:

a) $y = 3x^3 - x^2 - x - 1$ at $x = 0$
b) $y = x^4 + 3x^2 + x - 6$ at $x = 1$
c) $y = 5x^3 - x^2 + x$ at $x = -1$
d) $y = x^3 - 12x^2 + 48x - 2$ at $x = 4$

Question 5. The curve

$$y = x^6 - 3x^5 - 10x^4 + 10x^3 + 45x^2$$

is shown in the graph below.



Find all the points of inflection of the curve.

Question 6. Determine the values of x for which

$$y = 12x^5 - 20x^4 - 50x^3 + 90x^2$$

is convex (concave upwards). Give your answer in set notation.

Question 7. Find the coordinates of the points of inflection for each of the following curves. Determine in each case whether they are stationary or not.

a) $\mathrm{f}(x) = x^4 + 9x^3 - 15x^2 + 4$

b) $\mathrm{f}(x) = x^4 + 2x^3 - 2x - 1$

Question 8. Given that the function

$$h = 2t^3 - 9t^2 + 12t - 3$$

is used to model the height, h metres, of a passenger on a ride over time t, find the value of t when $\frac{d^2h}{dt^2} = 0$ and state the physical relevance of this point.

Question 9. The value of some shares £V during the first 12 months of the year may be modelled by

$$V = 0.25t^3 - 3.25t^2 + 10t + 20$$

where t is the number of months since the start of the year.

a) Use this model to predict the value of the shares when 12 months have passed.

b) Find when $\frac{d^2V}{dt^2} = 0$ and state its physical relevance.

Question 10. Show that

$$y = 2x^3 + 16x^2 + 43x + 38$$

has a single non-stationary point of inflection.

Question 11.

$$\mathrm{f}(x) = x^3 + 3x^2 + 3x + 2$$

a) Find the coordinates of the single stationary point of $y = \mathrm{f}(x)$
b) Show that the stationary point is a point of inflection of $y = \mathrm{f}(x)$
c) Hence, sketch the curve $y = \mathrm{f}(x)$

Question 12.

$$\mathrm{f}(x) = x^2 - 3x - \frac{1}{x}, \qquad x \neq 0$$

a) Find the stationary points of $y = \mathrm{f}(x)$ and determine their nature.
b) Show that there are no non-stationary points of inflection on the curve $y = \mathrm{f}(x)$

Question 13.

$$f(x) = 3x^4 - 4x^3 + 2$$

a) Show that the graph of $y = f(x)$ only has two stationary points.
b) Show that the two stationary points of $y = f(x)$ are different types of points.
c) Hence, sketch the curve $y = f(x)$

Question 14. For the curve $y = 2x^4 + 4x^3$

a) Find the stationary points on the curve and determine their type.
b) Find the non-stationary point of inflection.
c) Hence, sketch the curve.

Question 15. Find the values of a, b and c such that

$$y = x^3 + ax^2 + bx + c$$

has a single stationary point of inflection at the point $(5, 2)$

Question 16. Find the values of a, b and c such that

$$y = -x^3 + ax^2 + bx + c$$

has a single stationary point of inflection at the point $(1, 6)$

Question 17.

Prerequisite knowledge for Q17–Q18: Differentiation of the form ae^{kx} and solving equations involving e^x

Determine whether each of the curves are convex (concave upwards), concave (concave downwards) at the given point.

a) $y = e^x - x$ at $x = 2$
b) $y = 2e^x - e^{2x}$ at $x = 0$
c) $y = e^{2x} + 4$ at $x = -1$
d) $y = e^{-x} + x^2$ at $x = -1$
e) $y = e^{-x} - x^2$ at $x = 0$

Question 18.

a) Show that

$$y = e^x - x^2$$

has a single non-stationary point of inflection.
b) Find the coordinates of this point.

Question 19.

Prerequisite knowledge for Q19: Differentiation of Standard Functions

Show that there is a single non-stationary point of inflection on the curve

a) $y = x + \sin\left(\frac{1}{2}x\right), \qquad 0 < x \leq 2\pi$

b) $y = x^2 + \ln x, \qquad x > 0$

Question 20.

Prerequisite knowledge for Q20: Standard Functions and Product Rule

The price per litre (in pence), P, from January to June in 2024 is modelled by function

$$P = 144 + (t-3)e^{0.5t}$$

where t is the number of months after 1 January 2024.

a) According to the model, find the minimum price of the fuel in this time frame and the month in which it occurs.

b) Show that the curve has no points of inflection and explain what this means in context.